Object Storage Migration to Huawei Cloud OBS



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AI-Assisted Migration: Usage and Disclaimer

Sharing of capabilities, not a formal delivery commitment

Partner training | Huawei Cloud OMS/OBS for object storage migration

Central Note

This session presents the AI-assisted capabilities of Huawei Cloud for migrating object storage from AWS S3, Azure Blob, GCS, or on-prem to OBS. It does not process real customer data nor constitute a formal delivery plan, security or compliance commitment, or product feature promise.

1. Data and Privacy

Security / Compliance / Protection

- No real customer data is collected or processed
- Use simulated, test, or anonymized data
- Do not upload accounts, passwords, keys, certificates, or data

2. Usage Limits

Reference / Human Review

- AI supports planning, analysis, scripts, and risk assessment
- It does not replace design, expert review, testing, or decisions
- Review plans, scripts, and configurations before the project

3. Customer Authorization

Authorization / Approval

- The customer decides whether to use AI in real projects
- Without authorization, do not enter customer information into AI
- Respect security, compliance, confidentiality, and approvals

4. Not Recommended Scenarios

Prudent Evaluation / Manual Process

- Evaluate or avoid highly sensitive security scenarios
- Classified systems, financial core, defense, or energy
- Use manual migration under strict sovereignty or AI prohibition

AI-assisted migration improves efficiency; it does not replace human review or customer authorization. Real implementation must be based on customer confirmation, security, compliance, and project situation.

The Object Storage Migration Challenge



Source Environments

AWS S3, Azure Blob, GCS, Alibaba OSS, Tencent COS, on-prem NFS/MinIO. Typical SMB: 5-50TB across 10-50 buckets. Enterprise: 100TB-1PB across 100+ buckets.



Key Challenges

Data volume: TB to PB scale. Network bandwidth bottlenecks. Consistency during incremental sync. Metadata preservation. Lifecycle policy translation. Cost of downtime during cutover.



AI-Assisted Approach

AI scans source buckets, generates inventory. AI recommends OMS vs MgC based on volume. AI validates checksums post-migration. AI generates lifecycle rules for cost optimization.

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OMS & MgC Migration Workflow

OMS: Managed Migration



Console-based tool. Source: AK/SK + endpoint. Target: OBS bucket. Incremental sync. Filtering by prefix. 200-500 MB/s per task. 10TB: ~6-14 hours.

Validation & Consistency



Object count: source vs target. Checksum: MD5/SHA256 (automatic in OMS). Metadata: Content-Type, ACLs preserved. Sample test: 100 random objects. Report with success/failure counts.

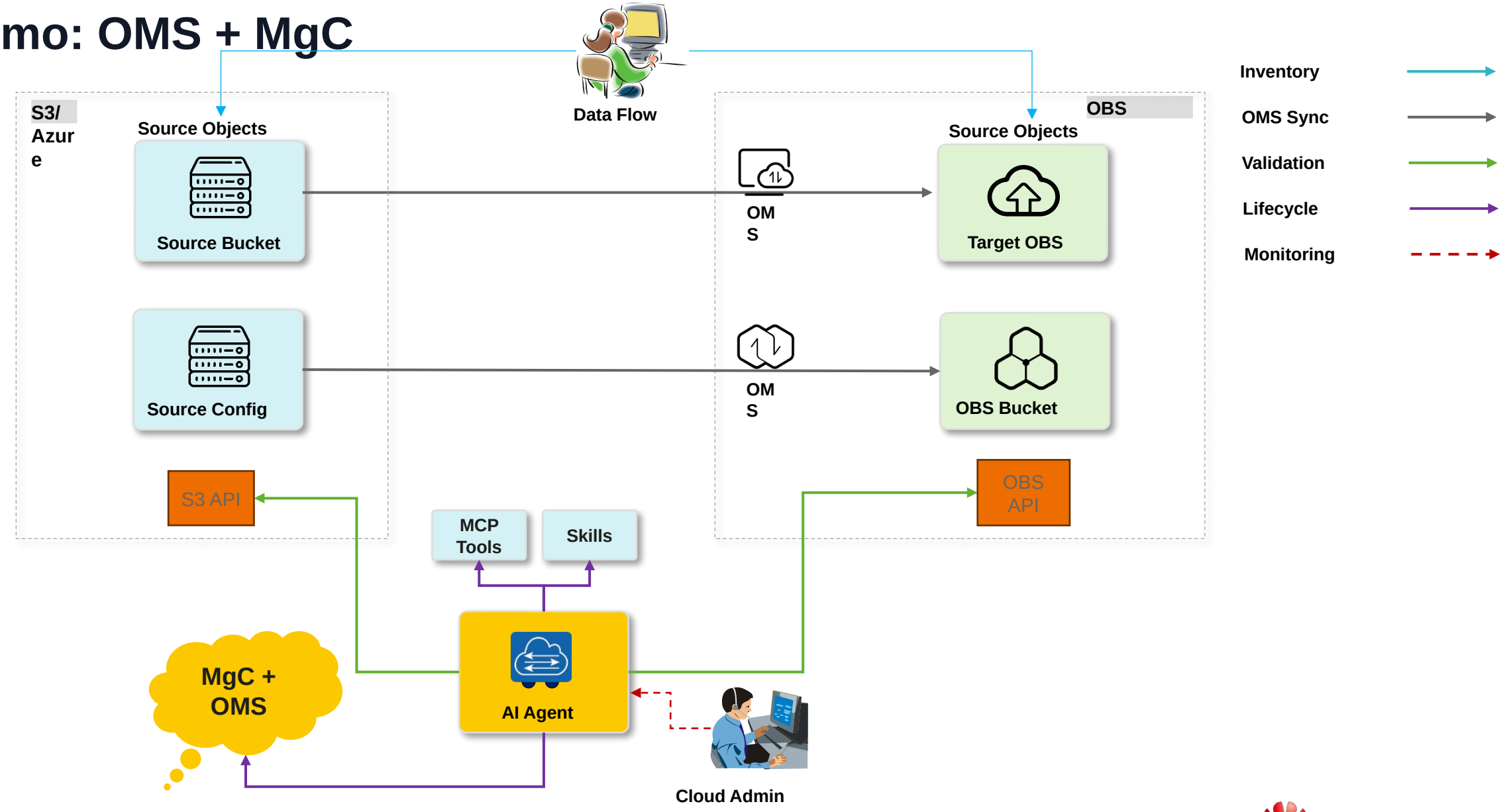
Large-Scale with MgC



For >100TB or multi-source. Batch: thousands of buckets in parallel. Pre-migration assessment. Checkpoint-based resume. Example: 500TB in ~10 days vs ~30 days sequential.

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Demo: OMS + MgC



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Source to Huawei Cloud Mapping

Source Cloud	Huawei Cloud	Tool	Notes
S3 (versioning)	OBS	OMS	Object storage migration
Azure Blob	OBS	OMS	Tier: Hot->Std, Cool->IA
GCS	OBS	OMS	S3-compatible API
Alibaba OSS	OBS	OMS	Region mapping required
On-prem NFS	OBS	MgC	Large-scale batch
MinIO	OBS	OMS	S3-compatible API
KMS	KMS	Manual	Re-encrypt with KMS keys
CloudWatch	CES	Manual	Monitoring setup
CloudFront	CDN	Manual	Edge caching config
S3 Lifecycle	OBS Lifecycle	Manual	Transition rules
—	LTS	Manual	Access logging

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Monitoring & Lifecycle

CES

Cloud Eye Monitoring

Features:
Request count, bandwidth, error rate
Storage usage and quota alerts
Access logs to LTS for audit
Cross-region replication status

Usage:
Monitor OBS bucket health
Alert on error spikes

Lifecycle

Auto-transition Rules

Features:
Hot -> IA (30d): 50% savings
IA -> Archive (90d): 80% savings
Archive -> Delete (365d)
Versioning + MFA delete

Usage:
10TB logs: $23->$11.5/mo
50% savings with lifecycle

CDN

Edge Caching

Features:
Cache at edge for lower latency
Reduce origin requests 70-90%
CDN egress < OBS egress
Pair with OBS for reads

Usage:
Read-heavy workloads
Global content distribution

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AI-Assisted Benefits

Operational Efficiency

AI scans source buckets, generates inventory.
AI recommends OMS vs MgC based on volume.
AI validates checksums post-migration.
AI generates lifecycle rules for cost optimization.

Cost Optimization & Risk

Lifecycle rules: 50-80% savings on infrequent data.
Right-tier: AI recommends IA/Archive.
CDN reduces egress costs for reads.
Checksum validation: zero data loss.
Incremental sync: zero downtime cutover.

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Resources & Next Steps

Resources

github.com/binrogithub/
1-3-Cloud-Adoption-Skills/
tree/main/AI-Assist-Migration/
Object-Migration

Next Steps

1. Inventory (Wk1): AI scans source.
2. Plan+test (Wk2): choose OMS/MgC.
3. Execute (Wk3-4): full migration.
4. Validate (Wk5): consistency check.
5. Cutover (Wk6): update configs.
6. Optimize (Wk7): lifecycle + CDN.

Contact

Huawei Cloud LATAM SMB team
for OMS/MgC migration assistance

OMS for single-source, MgC for large-scale multi-source.
AI scans, plans, validates, and optimizes.

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BENEFITS



Operational Efficiency

AI scans source buckets and generates inventory automatically. AI recommends OMS vs MgC based on volume. AI validates checksums and generates consistency reports.



Cost Optimization

Lifecycle rules: 50-80% savings on infrequent data. Right-tier: AI recommends IA/Archive. CDN reduces egress costs for read-heavy workloads.



Risk Reduction

Checksum validation: automatic data integrity. Incremental sync: zero data loss on cutover. Checkpoint-based resume on failure. 7-day source retention for rollback.

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Thank you.

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